LUMINOUS INTENSITY SENSOR

1. Product Overview

WIN-SN-LUX-200K-ME is a compact, high-precision Light intensity sensor designed for applications requiring low power consumption and small form factors. It ensures reliable measurements over a wide range. Its Modbus TCP communication interface simplifies integration into small, embedded systems, making it ideal for IOT. The sensor's long-term stability ensures consistent performance in harsh environments, making it a cost-effective solution for long-term deployment.

Fig.1 - Product Image

- Luminance to Digital Converter ranging 1 LUX to 200000 LUX.
- Superior sensor performance.
- Operating range -20°C to 60°C and 0% to 70% RH for humidity
- Fully calibrated and processed digital, Modbus TCP Output
- Wide Input voltage options 12-24V DC / 230V AC (optional)
- Excellent long-term stability

Measuring Parameter	Measuring Range	Operating Temperature
Luminous intensity	1-200000 LUX	-20°C ~ 60°C.

2. Sensor Specifications

- **Wide temperature range:** −40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*40 mm.
- **Certifications:** RoHS, CE APPROVED

3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don’t make loose connection.

4. Configuration Settings

- Communication IP 192.168.0.160
- CRC Yes
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

5. Modbus RS485 Data Storage register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Intensity (Lux)	40001 40002	0x03	TCP-IP	32 bit unsigned big endian
2	Lux range	40003 40004	0x03	TCP-IP	32 bit unsigned big endian
3	Read & Write Negative offset value	40005	0x03	TCP-IP	16-bit int
4	Read & Write Positive offset value	40006	0x03	TCP-IP	16-bit int

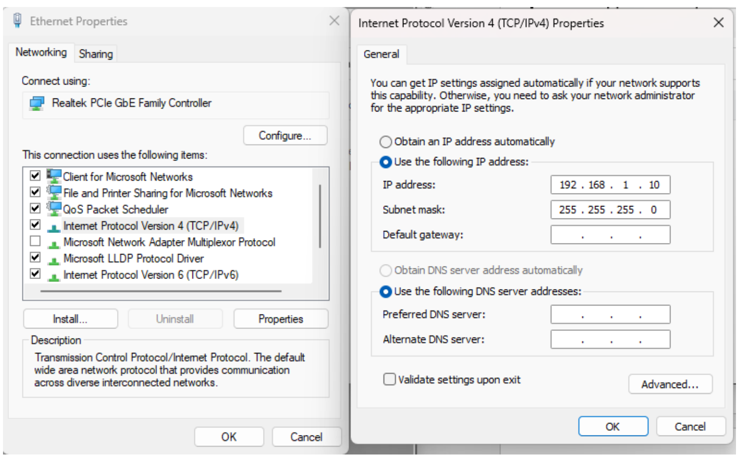
6. Modbus Configuration

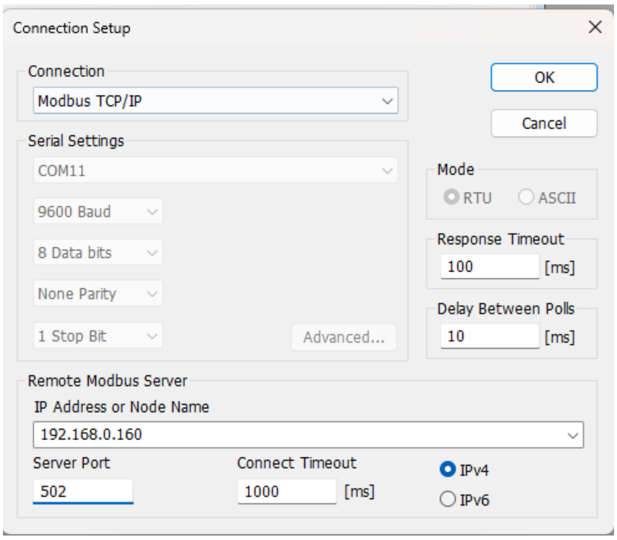
Here is an example of receiving data on Mb poll software.

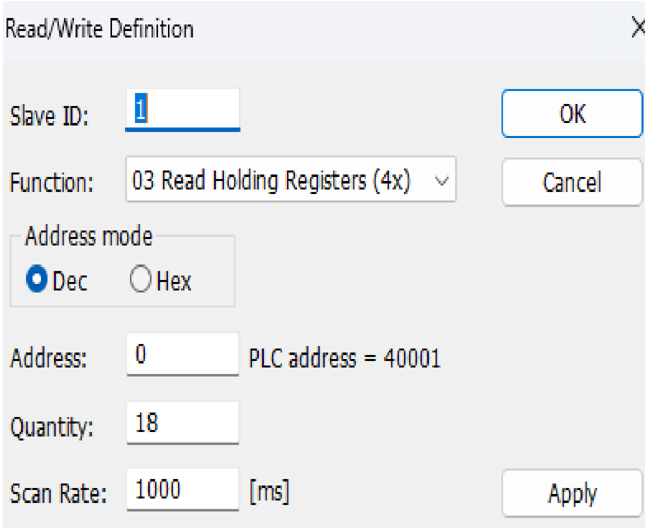
- Open Mb-poll software connect using configuration setting (Default TCP/IP 192.168.0.160).
- After connecting you will get data as shown below.

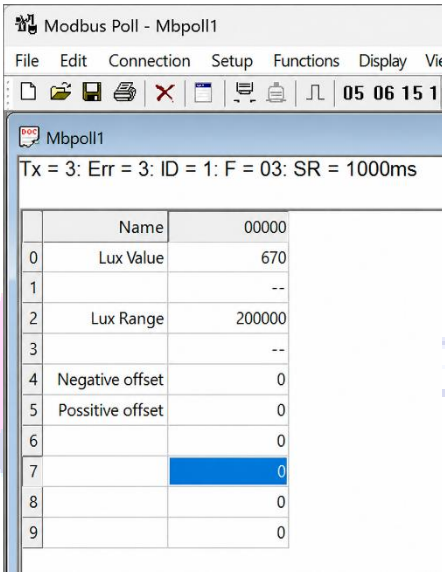
Important Note

Before connecting to the Modbus TCP network, ensure that the Ethernet IP address is configured in the **192.168.0.xx** subnet as shown below.

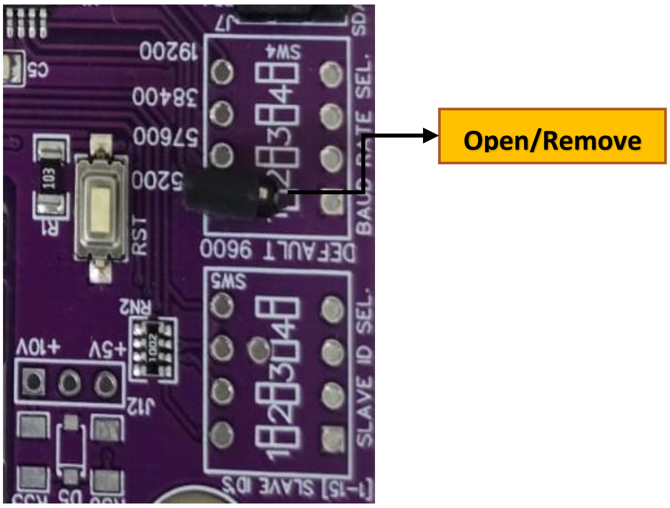








- Above image shows sensor data over Modbus with default Modbus TCP settings.
- Make the register 40001 40002 40003 40004 as 32 bit unsigned big endian then at register 40001 shows lux intensity & register 40003 shows lux Range.
- To set the maximum lux range, enter the desired maximum lux value in register 40003. The new value is applied immediately.
- To set the lux negative offset, enter the desired negative offset value in register 40004, to set the lux positive offset, enter the desired positive offset value in register 40005. The new value is applied immediately, and the updated lux value is reflected in register 40001.



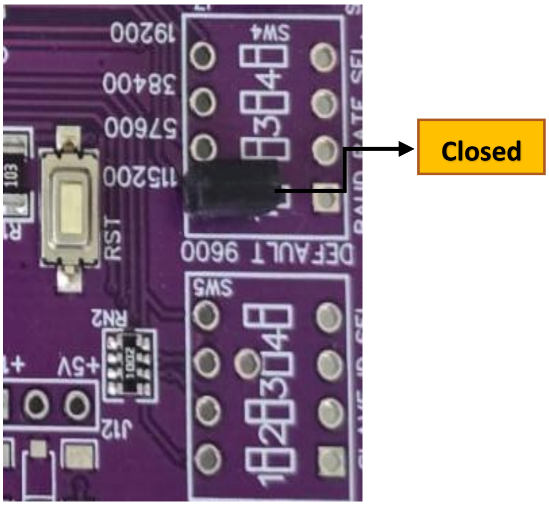


Image1.1 **Image1.2**

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

Contact Information

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